

EE-SE Pareto Boundary and Amplification Structure Optimization for Hybrid STAR-RIS

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Abstract—Hybrid active-passive simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) can improve spectral efficiency (SE) relative to fully passive architectures. Their energy efficiency (EE) behavior, however, remains insufficiently understood because existing analyses often rely on simplified amplifier-noise models, simplified control and switching power models, or asymmetric task models for the transmission and reflection sides. This paper develops a tractable analytical framework for characterizing the EE-SE Pareto boundary of hybrid active-passive STAR-RISs. We first introduce a three-part power model that accounts for static controller and bias power, dynamic switching overhead, and communication-side transmit and amplification power. For a sparse active set, we derive a closed-form SE-optimal amplification structure and an active-on threshold that quantifies when amplification gains offset amplifier noise. We then characterize the Pareto boundary via an ϵ -constraint formulation with a low-complexity sparse-activation solver and show that the resulting coefficient rules extend across ES, TS, and SS modes. Numerical results for the studied settings show that the proposed sparse-hybrid design improves EE over the baseline schemes.

Index Terms—STAR-RIS, active RIS, hybrid active-passive RIS, energy efficiency, spectral efficiency, Pareto boundary, amplifier noise.

I. INTRODUCTION

Simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) extend passive beam manipulation to simultaneous transmission and reflection, allowing a common surface to serve transmission-side and reflection-side users [2], [3]. This capability is attractive for asymmetric indoor-outdoor coverage. However, purely passive STAR-RIS still suffers from multiplicative path loss [1].

Sparse active elements can mitigate this loss at the price of amplifier noise and hardware power [4]–[7]. Measurement-driven studies further show that RIS power contains controller/bias and switching terms rather than a single constant [8], [9]. In STAR-RIS, these effects become more pronounced because the two sides compete for aperture, power split, and active hardware budget.

However, existing hybrid active-passive STAR-RIS studies do not clearly specify when the SE gain of an active element offsets its associated noise and power cost under given channel and hardware conditions [10]. Related active/passive STAR-RIS studies address neighboring settings such as SWIPT, URLLC, and ISAC [11]–[13]. To address this gap, we develop an equivalent formulation for hybrid STAR-RIS and derive tractable Pareto-boundary characterizations. The main contributions are as follows.

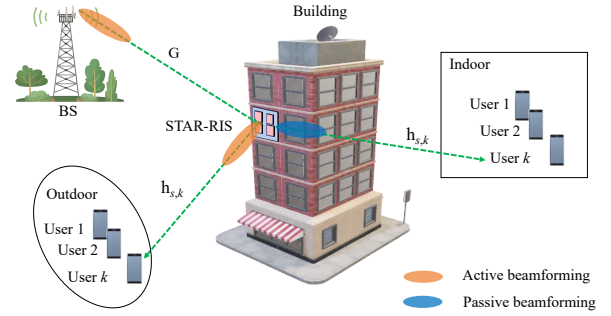


Fig. 1. The hybrid STAR-RIS-aided wireless system.

- We formulate a hybrid active-passive STAR-RIS downlink and introduce a three-part power model with BS transmit power, controller/bias power, switching overhead, and amplification power.
- For a fixed active set, we derive an equivalent side-wise SINR model, a closed-form SE-optimal common gain, an explicit active-on threshold, and a power-minimizing scalar refinement with a sufficient convexity condition.
- We develop a sparse-activation ϵ -constraint solver consistent with the analytical model and show that its coefficient rules extend across ES, TS, and SS. Simulations indicate that passive STAR-RIS is most competitive at low SE, whereas sparse hybrid activation yields higher EE at moderate-to-high SE under the considered settings.

II. SYSTEM MODEL

A. Hybrid active-passive STAR-RIS downlink

Consider a BS with M antennas serving K_t users on the transmission side and K_r users on the reflection side via an N -element STAR-RIS. Let $\mathcal{K}_t = \{1, \dots, K_t\}$ and $\mathcal{K}_r = \{1, \dots, K_r\}$. Under the ES protocol [3], the n th STAR-RIS element uses transmission and reflection coefficients

$$\theta_{n,s} = \sqrt{\rho_{n,s}} \beta_{n,s} e^{j\phi_{n,s}}, \quad s \in \{t, r\}, \quad (1)$$

with $\rho_{n,t} + \rho_{n,r} = 1$, $0 \leq \rho_{n,s} \leq 1$, and $1 \leq \beta_{n,s} \leq \beta_{\max}$. For passive elements, $\beta_{n,s} = 1$; for active elements, $\beta_{n,s} > 1$ on the assigned side. Let $\mathcal{A}_t$ and $\mathcal{A}_r$ denote the active-element sets for transmission and reflection, and let $\mathcal{A} = \mathcal{A}_t \cup \mathcal{A}_r$.

Denote by $\mathbf{G} \in \mathbb{C}^{N \times M}$ the BS-to-STAR-RIS channel, and by $\mathbf{h}_{s,k} \in \mathbb{C}^N$ the STAR-RIS-to-user channel on side s for user k . The received signal of user (s, k) is

$$y_{s,k} = \mathbf{h}_{s,k}^H \Theta_s \mathbf{G} \sum_{j \in \mathcal{K}_s} \mathbf{w}_{s,j} x_{s,j} + \mathbf{h}_{s,k}^H \mathbf{n}_{a,s} + n_{s,k}, \quad (2)$$

where $\Theta_s = \text{diag}(\theta_{1,s}, \dots, \theta_{N,s})$, $\mathbf{w}_{s,j}$ is the BS beamformer for user (s, j) , $x_{s,j}$ is the unit-power information symbol, and $n_{s,k} \sim \mathcal{CN}(0, \sigma_0^2)$ is thermal noise. The vector $\mathbf{n}_{a,s}$ collects the amplifier noises of the active elements on side s , and we model it as [4], [7]

$$\mathbf{n}_{a,s} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}_{a,s}), \quad \mathbf{C}_{a,s} = \text{diag}(c_{1,s}, \dots, c_{N,s}), \quad (3)$$

where $c_{n,s} = \rho_{n,s} \beta_{n,s}^2 \sigma_a^2$ for $n \in \mathcal{A}_s$ and $c_{n,s} = 0$ otherwise. The SINR of user (s, k) becomes

$$\gamma_{s,k} = \frac{|\mathbf{h}_{s,k}^H \Theta_s \mathbf{G} \mathbf{w}_{s,k}|^2}{\sum_{j \neq k} |\mathbf{h}_{s,k}^H \Theta_s \mathbf{G} \mathbf{w}_{s,j}|^2 + \sigma_{a,s,k}^2 + \sigma_0^2}, \quad (4)$$

where $\sigma_{a,s,k}^2 = \mathbf{h}_{s,k}^H \mathbf{C}_{a,s} \mathbf{h}_{s,k}$ is the active-noise term seen at user (s, k) .

B. Three-part power-consumption model

Motivated by recent measurement-based RIS power studies [8], [9], we write the total consumed power as

$$\begin{aligned} P_{\text{tot}} = & \frac{1}{\eta_{\text{PA}}} \sum_{s \in \{t,r\}} \sum_{k \in \mathcal{K}_s} \|\mathbf{w}_{s,k}\|^2 + P_{\text{ctrl}} + P_{\text{bias}} |\mathcal{A}| \\ & + f_u (P_{\text{sw},0} + P_{\text{sw},1} |\mathcal{A}|) \\ & + \sum_{n \in \mathcal{A}_t} \mu_t (\beta_{n,t}^2 - 1) + \sum_{n \in \mathcal{A}_r} \mu_r (\beta_{n,r}^2 - 1), \end{aligned} \quad (5)$$

where the terms represent BS transmit power, controller and bias power, switching overhead, and amplification power.

C. Structured equivalent formulation

To reveal the underlying structural properties, we introduce an equivalent formulation with side-wise zero-forcing directions, phase alignment on the dominant cascaded paths, and a common gain β_s for all active elements on side s . Let g_n be the effective BS-to-element channel after zero-forcing projection, and let $h_{s,k,n}$ be the n th entry of $\mathbf{h}_{s,k}$. We define the equivalent coherent-gain quantity

$$u_{s,k,n} = |g_n| |h_{s,k,n}|. \quad (6)$$

Then the side-wise effective coherent term for user (s, k) is

$$\psi_{s,k}(\beta_s) = \sum_{n \in \mathcal{P}_s} \sqrt{\rho_s} u_{s,k,n} + \beta_s \sum_{n \in \mathcal{A}_s} \sqrt{\rho_s} u_{s,k,n}, \quad (7)$$

where $\mathcal{P}_s = \{1, \dots, N\} \setminus \mathcal{A}_s$ contains passive elements and those active only for the opposite side. The corresponding active-noise term is

$$\zeta_{s,k}(\beta_s) = \sigma_0^2 + \rho_s \beta_s^2 \sigma_a^2 \sum_{n \in \mathcal{A}_s} |h_{s,k,n}|^2. \quad (8)$$

We further assume equal per-user rate splitting within each side. This captures side-wise asymmetric traffic demands but

not arbitrary user-specific QoS. The minimum sum transmit power needed to satisfy a side target rate R_s under equal user-rate splitting is then

$$p_s^{\min}(\beta_s) = \sum_{k \in \mathcal{K}_s} \frac{(2^{R_s/K_s} - 1) \zeta_{s,k}(\beta_s)}{\psi_{s,k}^2(\beta_s)}. \quad (9)$$

III. PROBLEM FORMULATION

The sum spectral efficiency is

$$R_\Sigma = \sum_{s \in \{t,r\}} \sum_{k \in \mathcal{K}_s} \log_2(1 + \gamma_{s,k}), \quad (10)$$

and the system energy efficiency is

$$\text{EE} = \frac{R_\Sigma}{P_{\text{tot}}} \quad (\text{bit/J}). \quad (11)$$

The original EE-maximization problem can be written as

$$\begin{aligned} \max_{\mathbf{w}, \rho, \phi, \beta, \mathcal{A}} \quad & \frac{R_\Sigma}{P_{\text{tot}}} \\ \text{s.t.} \quad & R_{s,k} \geq \bar{R}_{s,k}, \quad \forall s, k, \\ & \sum_{s,k} \|\mathbf{w}_{s,k}\|^2 \leq P_{\text{max}}^{\text{BS}}, \\ & 1 \leq \beta_{n,s} \leq \beta_{\text{max}}, \quad \forall n \in \mathcal{A}_s, \\ & \beta_{n,s} = 1, \quad \forall n \notin \mathcal{A}_s, \\ & \rho_{n,t} + \rho_{n,r} = 1, \quad 0 \leq \rho_{n,s} \leq 1, \\ & \rho_{n,s} \beta_{n,s}^2 \sigma_a^2 \leq \Gamma_{n,s}, \quad \forall n, s, \end{aligned} \quad (12)$$

where $\Gamma_{n,s}$ is a maximum noise-injection or hardware-safety limit. Problem (12) is highly non-convex. The fractional objective and the coupling among beamforming, STAR coefficients, active gains, and active-set selection make the problem difficult to solve directly.

To trace the EE-SE Pareto boundary, we adopt the ϵ -constraint formulation

$$\begin{aligned} \mathcal{P}(\epsilon) : \quad & \min_{\mathbf{w}, \rho, \phi, \beta, \mathcal{A}} \quad P_{\text{tot}} \\ \text{s.t.} \quad & R_\Sigma \geq \epsilon, \\ & \text{constraints of (12)}. \end{aligned} \quad (13)$$

Every feasible solution of (13) yields one Pareto-boundary point. We next exploit the model structure to derive transferable design rules.

IV. OPTIMAL STRUCTURE OF THE AMPLIFICATION FACTORS

A. Equivalent side-wise SINR law

For a fixed active set, define

$$\begin{aligned} a_s &= \min_k \sum_{n \in \mathcal{P}_s} \sqrt{\rho_s} u_{s,k,n}, \\ b_s &= \min_k \sum_{n \in \mathcal{A}_s} \sqrt{\rho_s} u_{s,k,n}, \\ c_s &= \sigma_0^2, \\ d_s &= \max_k \rho_s \sigma_a^2 \sum_{n \in \mathcal{A}_s} |h_{s,k,n}|^2. \end{aligned} \quad (14)$$

Here, a_s and b_s are the conservative worst-user coherent-gain aggregates from $\mathcal{P}_s$ and $\mathcal{A}_s$, while c_s and d_s denote the thermal-noise baseline and effective active-noise coefficient. With average side power $\bar{p}_s$, a conservative worst-user SINR lower bound is

$$\underline{\gamma}_s(\beta_s) = \frac{\bar{p}_s(a_s + b_s\beta_s)^2}{c_s + d_s\beta_s^2}. \quad (15)$$

Equation (15) captures the key tradeoff: coherent signal grows linearly in β_s , whereas injected active noise grows quadratically.

Proposition 1. *For fixed (a_s, b_s, c_s, d_s) with $a_s > 0$, $c_s > 0$, and $d_s > 0$, the function $\underline{\gamma}_s(\beta_s)$ in (15) is unimodal over $[1, \beta_{\max}]$, and its SE-optimal gain is*

$$\beta_{s,\text{SE}}^* = \left[\frac{b_s c_s}{a_s d_s} \right]_{[1, \beta_{\max}]}, \quad (16)$$

where $[x]_{[l,u]} = \min\{u, \max\{l, x\}\}$.

Proof: Differentiating (15) gives

$$\frac{d\underline{\gamma}_s}{d\beta_s} = \frac{2\bar{p}_s(a_s + b_s\beta_s)(b_s c_s - a_s d_s \beta_s)}{(c_s + d_s \beta_s^2)^2}. \quad (17)$$

The derivative changes sign only once, at $\beta_s = b_s c_s / (a_s d_s)$, which proves unimodality and (16). ■

A direct corollary of (17) is the *active-on threshold*:

$$\left. \frac{d\underline{\gamma}_s}{d\beta_s} \right|_{\beta_s=1} > 0 \iff b_s c_s > a_s d_s \iff \frac{b_s}{a_s} > \frac{d_s}{c_s}. \quad (18)$$

Equation (18) specifies when active amplification is beneficial: the coherent-gain increase must exceed the normalized amplifier-noise penalty. At high SNR (small effective c_s), this condition becomes harder to satisfy.

B. Power-minimizing gain and a sufficient convexity condition

For a target side rate R_s , we define the cost as the sum of the minimum BS transmit power required to meet the rate target and the intrinsic power consumed by the active amplifiers:

$$\Psi_s(\beta_s) = \frac{K_s(2^{R_s/K_s} - 1)(c_s + d_s\beta_s^2)}{\eta_{\text{PA}}(a_s + b_s\beta_s)^2} + \mu_s |\mathcal{A}_s| (\beta_s^2 - 1). \quad (19)$$

Its first derivative is

$$\Psi'_s(\beta_s) = \frac{2K_s(2^{R_s/K_s} - 1)(a_s d_s \beta_s - b_s c_s)}{\eta_{\text{PA}}(a_s + b_s\beta_s)^3} + 2\mu_s |\mathcal{A}_s| \beta_s, \quad (20)$$

and its second derivative is

$$\begin{aligned} \Psi''_s(\beta_s) &= \frac{2K_s(2^{R_s/K_s} - 1)}{\eta_{\text{PA}}(a_s + b_s\beta_s)^4} (a_s^2 d_s - 2a_s b_s d_s \beta_s + 3b_s^2 c_s) \\ &\quad + 2\mu_s |\mathcal{A}_s|. \end{aligned} \quad (21)$$

Proposition 2. *Assume the mixed case $a_s > 0$, $b_s > 0$, and $d_s > 0$. If*

$$\beta_{\max} \leq \frac{a_s^2 d_s + 3b_s^2 c_s}{2a_s b_s d_s}, \quad (22)$$

TABLE I
CASE CLASSIFICATION FOR THE OPTIMIZATION OF β_s

Case	Condition	Conclusion
Fully passive	$ \mathcal{A}_s = 0$	$b_s = d_s = 0$; no active gain is to be optimized; set $\beta_s^* = 1$.
Active-no-benefit degenerate case	$ \mathcal{A}_s > 0$, $a_s > 0$, $b_s = 0$	$\Psi'_s(\beta_s) > 0$; $\Psi_s(\beta_s)$ is strictly increasing; set $\beta_s^* = 1$.
Fully active	$ \mathcal{A}_s > 0$, $a_s = 0$, $b_s > 0$	$\Psi''_s(\beta_s) > 0$; $\Psi_s(\beta_s)$ is strictly convex and admits a unique minimizer.
Mixed active-passive	$a_s > 0$, $b_s > 0$	If $\beta_{\max} \leq \frac{a_s^2 d_s + 3b_s^2 c_s}{2a_s b_s d_s}$, then $\Psi_s(\beta_s)$ is strictly convex on $[1, \beta_{\max}]$ and the minimizer is unique.

then $\Psi_s(\beta_s)$ is strictly convex on $[1, \beta_{\max}]$. Consequently, the side-wise power-minimizing gain is unique. It can be found by checking the two boundaries and the unique root of (20).

Proof: Under (22), the bracketed term in (21) is non-negative on $[1, \beta_{\max}]$ and, being affine in β_s , can vanish at most once. Hence $\Psi''_s(\beta_s)$ is non-negative everywhere and positive on every nontrivial subinterval. Therefore, $\Psi'_s(\beta_s)$ is strictly increasing and $\Psi_s(\beta_s)$ is strictly convex. The unique minimum is attained at an endpoint or at the unique root of (20). ■

In the numerical study, we therefore set $\beta_s = 1$ in the first two cases. We solve the strictly convex scalar problem in the fully active case and use (16) with the refinement of Proposition 2 in the mixed case.

C. Extension of the coefficient rules to TS and SS

Since ES is already covered by (14), for $p \in \{\text{TS}, \text{SS}\}$ let $\mathcal{P}_s^{(p)}$ and $\mathcal{A}_s^{(p)}$ denote the passive and active support sets. We then define

$$\begin{aligned} a_s^{(p)} &= \min_k \sum_{n \in \mathcal{P}_s^{(p)}} u_{s,k,n}, & b_s^{(p)} &= \min_k \sum_{n \in \mathcal{A}_s^{(p)}} u_{s,k,n}, \\ c_s^{(p)} &= \sigma_0^2, & d_s^{(p)} &= \max_k \sigma_a^2 \sum_{n \in \mathcal{A}_s^{(p)}} |h_{s,k,n}|^2. \end{aligned}$$

For TS, $\mathcal{P}_s^{(\text{TS})} = \{1, \dots, N\} \setminus \mathcal{A}_s$ and the effective target becomes R_s/τ_s . For SS, $\mathcal{P}_s^{(\text{SS})} = \mathcal{S}_s \setminus \mathcal{A}_s$ with $\mathcal{S}_t \cup \mathcal{S}_r = \{1, \dots, N\}$. Thus (18) applies after replacing (a_s, b_s, c_s, d_s) by $(a_s^{(p)}, b_s^{(p)}, c_s^{(p)}, d_s^{(p)})$.

V. SPARSE-ACTIVATION PARETO BOUNDARY ALGORITHM

A. Outer sparse active-element selection

Motivated by the active-on threshold in (18), we propose a heuristic sparse-activation strategy. It prioritizes elements offering a high ratio of coherent signal gain to noise-and-hardware cost. For a candidate element $n \notin \mathcal{A}_s$, we introduce a relaxed inclusion variable $x_{n,s} \in [0, 1]$. For fixed R_s , β_s , and current aggregates (a_s, b_s, c_s, d_s) , define

$$C_s = \frac{K_s(2^{R_s/K_s} - 1)}{\eta_{\text{PA}}}, \quad \Delta u_{n,s} = \sqrt{\rho_s} |g_n| \min_k |h_{s,k,n}|.$$

The exact increment of d_s depends on the maximizing user after inserting element n ; for ranking we use the surrogate

$$\Delta d_{n,s} \approx \rho_s \sigma_a^2 \overline{|h_{s,k,n}|^2}, \quad \overline{|h_{s,k,n}|^2} \triangleq \frac{1}{K_s} \sum_{k \in \mathcal{K}_s} |h_{s,k,n}|^2.$$

Under the same conservative surrogate used in (15)–(19), the n -dependent part of the side cost is approximated by

$$\begin{aligned} \Psi_s(x_{n,s}) \approx & C_s \frac{c_s + \beta_s^2(d_s + x_{n,s}\Delta d_{n,s})}{(a_s + b_s\beta_s + (\beta_s - 1)x_{n,s}\Delta u_{n,s})^2} \\ & + \mu_s(|\mathcal{A}_s| + x_{n,s})(\beta_s^2 - 1) \\ & + x_{n,s}(P_{\text{bias}} + f_u P_{\text{sw},1}). \end{aligned}$$

Therefore, the first-order increment at $x_{n,s} = 0$ is

$$\begin{aligned} \left. \frac{\partial \Psi_s}{\partial x_{n,s}} \right|_{x_{n,s}=0} = & -\kappa_{1,s}\Delta u_{n,s} + \kappa_{2,s}\Delta d_{n,s} \\ & + P_{\text{bias}} + f_u P_{\text{sw},1} + \mu_s(\beta_s^2 - 1), \end{aligned}$$

where

$$\kappa_{1,s} = \frac{2C_s(c_s + d_s\beta_s^2)(\beta_s - 1)}{(a_s + b_s\beta_s)^3}, \quad \kappa_{2,s} = \frac{C_s\beta_s^2}{(a_s + b_s\beta_s)^2}.$$

Hence first-order activation is beneficial whenever

$$\frac{\kappa_{1,s}\Delta u_{n,s}}{\kappa_{2,s}\Delta d_{n,s} + P_{\text{bias}} + f_u P_{\text{sw},1} + \mu_s(\beta_s^2 - 1)} > 1.$$

Absorbing $\kappa_{1,s}$ and $\kappa_{2,s}$ into normalized weights yields the practical ranking score

$$\xi_{n,s} = \frac{\omega_s \sqrt{\rho_s} |g_n| \min_k |h_{s,k,n}|}{\lambda_1 \rho_s \sigma_a^2 \overline{|h_{s,k,n}|^2} + \lambda_2 (P_{\text{bias}} + f_u P_{\text{sw},1})}, \quad (23)$$

where ω_s reflects the side demand and λ_1, λ_2 absorb the side-wise constants. In the simulations, $\omega_t = \alpha$, $\omega_r = 1 - \alpha$, $\lambda_1 = \sigma_0^{-2}$, and $\lambda_2 = (P_{\text{max}}^{\text{BS}})^{-1}$. Candidates are ranked by $\max_s \xi_{n,s}$ and assigned to the maximizing side. They are then re-evaluated through (9), (19), and (5).

B. Inner gain and rate-split update

For a given active cardinality $L = |\mathcal{A}|$, we search over the 1-D rate split $R_t = \alpha\epsilon$ and $R_r = (1 - \alpha)\epsilon$. After updating $\mathcal{A}_t$ and $\mathcal{A}_r$ via (23), we compute (a_s, b_s, c_s, d_s) and update β_s case by case. We set $\beta_s = 1$ if $|\mathcal{A}_s| = 0$ or $b_s = 0$, solve $\min_{\beta_s \in [1, \beta_{\text{max}}]} \Psi_s(\beta_s)$ if $a_s = 0$, and otherwise use (16) or Proposition 2. The best feasible (L, α) minimizes (5).

C. Convergence and complexity

Algorithm 1 terminates in finitely many steps because $\mathcal{L}$ and $\mathcal{G}_\alpha$ are finite. With closed-form gain updates, the dominant per-boundary-point complexity is

$$\mathcal{O}(|\mathcal{L}||\mathcal{G}_\alpha|(N \log N + N(K_t + K_r))),$$

where the $N \log N$ term comes from ranking the candidate active elements; adding the scalar refinement of Proposition 2 only introduces a multiplicative 1-D search factor.

Algorithm 1 Sparse-Activation Solver

Require: Target ϵ , cap β_{max} , grids $\mathcal{L}$ and $\mathcal{G}_\alpha$

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1: Initialize best power as  $+\infty$ 
2: for  $L \in \mathcal{L}$  do
3:   for  $\alpha \in \mathcal{G}_\alpha$  do
4:     Build  $\mathcal{A}_t, \mathcal{A}_r$  using score (23)
5:     Compute  $(a_s, b_s, c_s, d_s)$  for  $s \in \{t, r\}$ 
6:     for all  $s \in \{t, r\}$  do
7:       if  $|\mathcal{A}_s| = 0$  or  $b_s = 0$  then
8:         Set  $\beta_s \leftarrow 1$ 
9:       else if  $a_s = 0$  then
10:        Minimize  $\Psi_s(\beta_s)$  on  $[1, \beta_{\text{max}}]$ 
11:      else
12:        Update  $\beta_s$  using (16) or Proposition 2
13:      end if
14:    end for
15:    Compute the required BS power via (9)
16:    if  $R_\Sigma \geq \epsilon$  and all constraints hold then
17:      Evaluate  $P_{\text{tot}}$  from (5)
18:      Keep the candidate with minimum  $P_{\text{tot}}$ 
19:    end if
20:  end for
21: end for
22: return best feasible candidate

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TABLE II
SIMULATION PARAMETERS

Parameter	Value
STAR-RIS elements N	24
Users (K_t, K_r)	(2, 2)
Fading model	Nakagami- m
Nakagami parameters (m_{BR}, m_{RU})	(1.0, 1.0)
ES coefficients (ρ_t, ρ_r)	(0.55, 0.45)
TS slot fractions (τ_t, τ_r)	(0.55, 0.45)
Noise std. (σ_0, σ_a)	(0.26, 0.05)
Maximum active gain β_{max}	5
BS power budget $P_{\text{max}}^{\text{BS}}$	5.0 W
Static controller power P_{ctrl}	0.12 W
Bias power P_{bias}	0.016 W per element
Dynamic base and slope $(P_{\text{sw},0}, P_{\text{sw},1})$	(0.01, 0.006) W
Amplifier coefficient $\mu_t = \mu_r$	0.0025 W
PA efficiency η_{PA}	0.38
Monte Carlo trials	100

VI. NUMERICAL RESULTS

We consider a hybrid STAR-RIS with $N = 24$ elements, $K_t = K_r = 2$, Nakagami- m fading, and the normalized parameters listed in Table II. Each curve averages 100 independent channel draws. Unless stated otherwise, the BS power budget is 5.0 W. The ES coefficients are $\rho_t = 0.55$ and $\rho_r = 0.45$, the TS slot fractions are $\tau_t = 0.55$ and $\tau_r = 0.45$, and the gain cap is $\beta_{\text{max}} = 5$.

Fig. 2 shows the EE-SE Pareto boundary. Passive STAR-RIS is most competitive at very low SE, whereas the fully active benchmark incurs higher amplification cost as the target SE increases. Among the considered schemes, the proposed sparse-hybrid design achieves the highest EE in the medium-to-high-SE region and remains feasible up to 20 bit/s/Hz. At 12 bit/s/Hz, the EE gain is approximately 30.7% over uniform

hybrid activation, 74.9% over the fully active design, and 94.4% over the passive design.

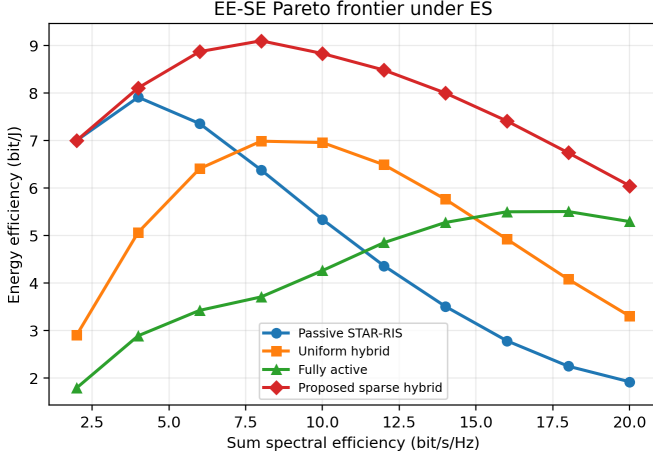


Fig. 2. EE-SE Pareto boundary under ES.

Fig. 3 compares the cross-mode design implications. TS requires the densest active set because the effective slot targets are inflated by $1/\tau_s$. At 12 bit/s/Hz, its active ratio is about 0.54, versus about 0.27 for ES and 0.29 for SS. SS therefore tracks ES more closely than TS in activation density. The gain curves indicate that SS stays near the gain cap, TS also uses high gains but with denser activation, and ES increases more gradually before saturation.

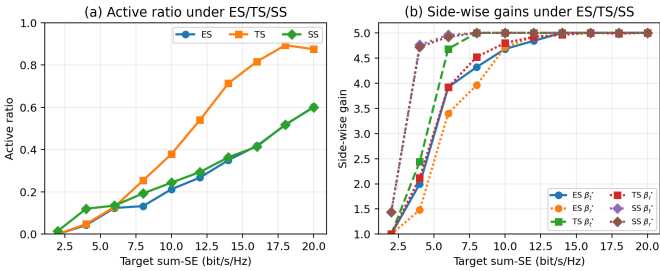


Fig. 3. Cross-mode design rules across ES, TS, and SS: (a) active ratio; (b) side-wise common gain.

Fig. 4 shows the hardware sensitivity across ES, TS, and SS at $R_\Sigma = 12$ bit/s/Hz. In Fig. 4(a), the average side-wise gain decreases with σ_a for all three protocols. The drop is largest under TS because slot-rate inflation already forces denser activation. In Fig. 4(b), increasing the update frequency from 50 Hz to 550 Hz reduces EE for all three protocols. SS stays slightly above ES, whereas TS remains lower because its denser activation pattern magnifies the dynamic-cost penalty. The framework captures side-wise asymmetric rate demands through (a_s, b_s, d_s) and the rate split α . Heterogeneous user-specific QoS and joint calibration/forwarding extensions are left for future work.

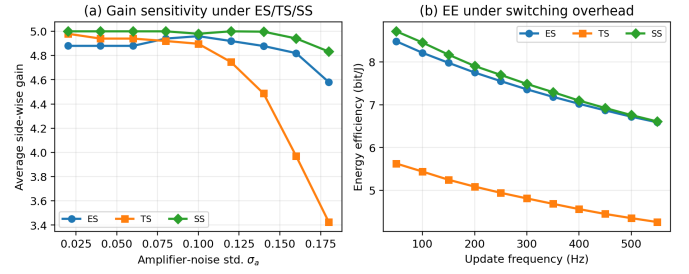


Fig. 4. Hardware sensitivity across ES, TS, and SS: (a) average side-wise gain versus amplifier noise; (b) EE versus switching overhead.

Fig. 5 compares the ES power decomposition at $R_\Sigma = 12$ bit/s/Hz. The fully active design minimizes the BS transmit-power share, about 0.38 W on average, but also incurs the largest static/dynamic and amplification costs, with the amplification term alone reaching about 1.44 W. The passive design avoids amplification but requires about 2.78 W of BS power. The proposed sparse-hybrid scheme achieves the lowest total power, about 1.44 W versus about 1.91 W for uniform hybrid, by reducing the BS radiated-power requirement while keeping the amplification term well below the fully active benchmark.

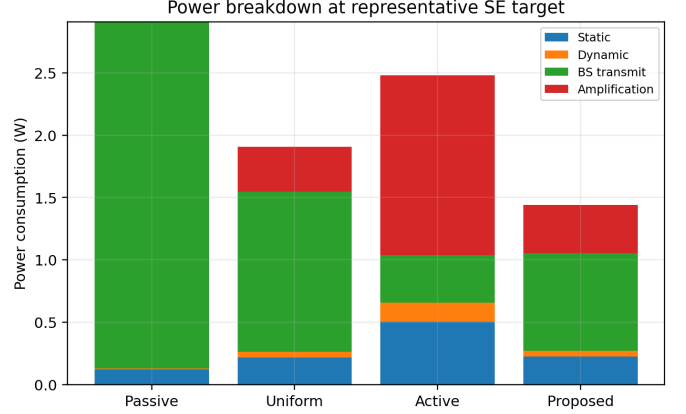


Fig. 5. Power breakdown at $R_\Sigma = 12$ bit/s/Hz under ES.

VII. CONCLUSION

This paper studied the EE-SE tradeoff of hybrid active-passive STAR-RIS through a structured Pareto-boundary formulation linking amplifier noise, active-set sparsity, side-wise asymmetry, and hardware power. The analysis yielded a closed-form SE-optimal gain, an active-on threshold, and a sparse-activation rule. The ϵ -constraint solver translated these results into a reproducible boundary generator. Numerical results indicated that passive STAR-RIS is most competitive at low SE, whereas sparse hybrid activation yields higher EE at moderate-to-high SE under the considered settings. TS requires denser activation and is more sensitive to amplifier noise than ES and SS. Extending the framework to CSI-imperfect or calibration-aware designs remains future work.

REFERENCES

- [1] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.
- [2] Y. Liu, X. Mu, J. Xu, R. Schober, Y. Hao, H. V. Poor, and L. Hanzo, "STAR: Simultaneous transmission and reflection for 360° coverage by intelligent surfaces," *IEEE Wireless Commun.*, vol. 28, no. 6, pp. 102–109, Dec. 2021, doi: 10.1109/MWC.001.2100191.

- [3] X. Mu, Y. Liu, L. Guo, J. Lin, and R. Schober, "Simultaneously transmitting and reflecting (STAR) RIS aided wireless communications," *IEEE Trans. Wireless Commun.*, vol. 21, no. 5, pp. 3083–3098, May 2022.
- [4] Y. Ma, M. Li, Y. Liu, Q. Wu, and Q. Liu, "Active reconfigurable intelligent surface for energy efficiency in MU-MISO systems," *IEEE Trans. Veh. Technol.*, vol. 72, no. 3, pp. 4103–4107, Mar. 2023.
- [5] Q. Peng, Q. Wu, G. Chen, R. Liu, S. Ma, and W. Chen, "Hybrid active-passive IRS assisted energy-efficient wireless communication," *IEEE Commun. Lett.*, vol. 27, no. 8, pp. 2202–2206, Aug. 2023.
- [6] A. Huang, X. Mu, L. Guo, and G. Zhu, "Hybrid active-passive RIS transmitter enabled energy-efficient multi-user communications," *IEEE Trans. Wireless Commun.*, vol. 23, no. 9, pp. 10653–10666, Sep. 2024.
- [7] Z. Zhang, L. Dai, X. Chen, C. Liu, F. Yang, R. Schober, and H. V. Poor, "Active RISs: Signal modeling, asymptotic analysis, and beamforming design," in *Proc. IEEE GLOBECOM*, Rio de Janeiro, Brazil, 2022, pp. 1618–1624, doi: 10.1109/GLOBECOM48099.2022.10001687.
- [8] J. Wang, W. Tang, S. Jin, X. Li, and M. Matthaiou, "Static power consumption modeling and measurement of reconfigurable intelligent surfaces," in *Proc. EUSIPCO*, Helsinki, Finland, 2023, pp. 890–894.
- [9] J. Wang, W. Tang, J. C. Liang, L. Zhang, J. Y. Dai, X. Li, S. Jin, Q. Cheng, and T. J. Cui, "Reconfigurable intelligent surface: Power consumption modeling and practical measurement validation," *IEEE Trans. Commun.*, vol. 72, no. 9, pp. 5720–5734, Sep. 2024, doi: 10.1109/TCOMM.2024.3382332.
- [10] T.-H. Vu, T. N. Nguyen, T.-T. Nguyen, and S. Kim, "Hybrid active-passive STAR-RIS-based NOMA systems: Energy/rate-reliability trade-offs and rate adaptation," *IEEE Wireless Commun. Lett.*, vol. 14, no. 1, pp. 238–242, Jan. 2025.
- [11] G. Zhu, X. Mu, L. Guo, A. Huang, and S. Xu, "STAR-RIS assisted SWIPT systems: Active or passive?" *IEEE Trans. Wireless Commun.*, vol. 25, pp. 4514–4529, 2026, doi: 10.1109/TWC.2025.3611793.
- [12] M. Soleymani, I. Santamaria, and E. A. Jorswieck, "Spectral and energy efficiency maximization of MISO STAR-RIS-assisted URLLC systems," *IEEE Access*, vol. 11, pp. 70833–70852, 2023.
- [13] S. Zhang, W. Hao, G. Sun, C. Huang, Z. Zhu, X. Li, and C. Yuen, "Joint beamforming optimization for active STAR-RIS-assisted ISAC systems," *IEEE Trans. Wireless Commun.*, vol. 23, no. 11, pp. 15888–15902, Nov. 2024.